Sahaj Bhadja

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Summary

IT applications engineer with 2+ years designing, developing, maintaining, monitoring, testing, and supporting **business-critical software and billing workflows**; experienced with SQL, Claude, Agile, Jira, automation, CI/CD, and production troubleshooting.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Analysis of Algorithms, Database Systems, Web Technologies, Advanced NLP

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Software Engineering, Operating Systems, Object-Oriented Programming, Data Retrieval

Experience

USC Dornsife **Los Angeles, CA**
Software Engineer *September 2025 – Present*

- Designed and supported scalable, business-critical production applications using Python, FastAPI, PostgreSQL, and SQL Server, serving 500+ users while reducing dashboard latency from 1.2 seconds to 350 milliseconds.
- Monitored application health and troubleshot 5,000+ recurring monthly integration workflows using reconciliation, error handling, validation, and CI/CD release checks, sustaining 99%+ reliability across integrated production systems.

Grad8 **Los Angeles, CA**
Software Engineer Intern *June 2025 – August 2025*

- Built AI-assisted application features using Claude 3.7, Python, React, and AWS, serving 150+ users and 10,000+ daily production requests while maintaining stable performance across operational workflows.
- Developed robust automated test processes and validation strategies for file ingestion, grading, and model fallbacks, reducing feedback turnaround 60% while sustaining 90%+ consistency across 10,000+ submissions.

Aasma Technology Solutions **India**
Software Developer Engineer - (SecureSync) *August 2023 – July 2024*

- Maintained and debugged complex cross-platform applications and integrations using SQL, AWS Glue, and S3, reducing response times 30%, technical debt 20%, and data-processing overhead across distributed systems.
- Created 20+ automated test scripts and CI/CD workflows across Linux, Windows, and macOS, documenting failure metrics and improving software build and release efficiency 68% through Agile delivery.

Aasma Technology Solutions **India**
Full Stack Developer Intern - (VibrantChem) *May 2023 – August 2023*

- Implemented billing, customer, permissions, and notification services backed by Azure SQL, synchronizing two million records through REST APIs and WebSockets while improving maintainability across eight modular applications.
- Configured monitoring with Sentry, Azure Monitor, and Kibana and automated testing and deployment through Jenkins, achieving 95%+ build validation while reducing release time 50% and review turnaround 25%.

LST Capitals **India**
Web Developer Intern *November 2022 – January 2023*

- Developed complex secure booking, pricing, payment, and refund workflows using Python, Flask, and Firebase, supporting 50,000+ records with sub-200-millisecond responses and reducing manual payment handling 70%.
- Participated in requirements analysis and technical design with customers, documenting 12+ integration flows and resolving operational discrepancies across mobile and web applications while reducing design errors 25%.

Skills

Applications: Requirements analysis, technical specifications, system configuration, maintenance, support, troubleshooting
Data: SQL, SQL Server, Azure SQL, PostgreSQL, MySQL, MongoDB, schema design, reconciliation
Automation: Python, Shell, Jenkins, GitHub Actions, CI/CD, build automation, deployment scripting, release management
Quality: Test plans, automated test scripts, unit testing, integration testing, validation, debugging, quality assurance
Operations: Claude, Agile, Scrum, Jira, Git, Sentry, Azure Monitor, Kibana, application health, technical risks

Projects

HMIS Live | *Healthcare Workflows, SQL, Monitoring*

- Built a healthcare application with consent-based patient access, role-based workflows, REST integrations, and Sentry monitoring, supporting 100+ daily appointments while reducing record retrieval time 50% and administration 30%.

Chat2SQL Code | *SQL, Validation, Database Management*

- Extended an AI-assisted SQL management platform supporting 10+ databases, implementing schema exploration, robust query validation, structured execution, and cross-platform debugging through a Java backend and TypeScript frontend.

FinNexus Live | *Billing, CI/CD, Application Monitoring*

- Developed secure billing and transaction workflows supporting 50,000+ users and one million records, combining real-time updates, Kubernetes deployment automation, application monitoring, and scalable MongoDB-backed data access.